KAIZEN MATHS

Lesson starter

Function Notation, Composite and Inverse Functions

Expression Inputs and Composition

Name

S1

For f(x) = 2x + 8, find f(4x + 5).

S2

For $f(x) = -x^2 + x + 1$, find f(3x).

S3

For f(x) = -x, g(x) = -2x - 5, and h(x) = -x + 5, find f(g(h(x))).

S4

For f(x) = -x + 7, find f(6x - 6).

Show your working

KAIZEN MATHS

Lesson starter

Function Notation, Composite and Inverse Functions

Expression Inputs and Composition

Name

S1

For f(x) = 2x + 8, find f(4x + 5).

S2

For $f(x) = -x^2 + x + 1$, find f(3x).

S3

For f(x) = -x, g(x) = -2x - 5, and h(x) = -x + 5, find f(g(h(x))).

S4

For f(x) = -x + 7, find f(6x - 6).

ALG-2F88-L2 · S

Show your working

KAIZEN MATHS

Exit ticket

Function Notation, Composite and Inverse Functions

Expression Inputs and Composition

Name _____

E1

For $f(x) = x^2 - x - 3$, find $f(-2x - 3)$.

E2

For $f(x) = x + 3$, $g(x) = 5x - 2$, and $h(x) = 3x + 4$, find $f(g(h(x)))$.

E3

For $f(x) = -3x - 8$, find $f(-2x + 6)$.

KAIZEN MATHS

Exit ticket

Function Notation, Composite and Inverse Functions

Expression Inputs and Composition

Name _____

E1

For $f(x) = x^2 - x - 3$, find $f(-2x - 3)$.

E2

For $f(x) = x + 3$, $g(x) = 5x - 2$, and $h(x) = 3x + 4$, find $f(g(h(x)))$.

E3

For $f(x) = -3x - 8$, find $f(-2x + 6)$.

ALG-2F88-L2 · E

Show your working

KAIZEN MATHS

Exit ticket

Function Notation, Composite and Inverse Functions

Expression Inputs and Composition

Name _____

E1

For $f(x) = x^2 - x - 3$, find $f(-2x - 3)$.

E2

For $f(x) = x + 3$, $g(x) = 5x - 2$, and $h(x) = 3x + 4$, find $f(g(h(x)))$.

E3

For $f(x) = -3x - 8$, find $f(-2x + 6)$.

ALG-2F88-L2 · E

Show your working

Teacher answers & working

Function Notation, Composite and Inverse Functions · Expression Inputs and Composition

ALG-2F88-L2 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

For f(x) = 2x + 8, find f(4x + 5).

Answer:

$8x + 18$

Working:

Step 1: Treat 4x + 5 as one complete input and substitute it for x.

$f(4x + 5) = 2(4x + 5) + 8$

Step 2: Expand the bracket.

$f(4x + 5) = 8x + 10 + 8$

Step 3: Collect the constant terms.

$f(4x + 5) = 8x + 18$

S2

For $f(x) = -x^2 + x + 1$, find f(3x).

Answer:

$-9x^2 + 3x + 1$

Working:

Step 1: Substitute the whole expression 3x wherever x appears.

$f(3x) = -1(3x)^2 + 1(3x) + 1$

Step 2: Expand the squared bracket and the linear bracket.

$f(3x) = -9x^2 + 1(3x) + 1$

Step 3: Collect like powers of x.

$f(3x) = -9x^2 + 3x + 1$

Teacher answers & working

Function Notation, Composite and Inverse Functions · Expression Inputs and Composition

ALG-2F88-L2 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S3

For f(x) = -x, g(x) = -2x - 5, and h(x) = -x + 5, find f(g(h(x))).

Answer:

$-2x + 15$

Working:

Step 1: Start with the innermost function, h(x).

$h(x) = -x + 5$

Step 2: Substitute h(x) into g(x).

$g(h(x)) = -2(-x + 5) - 5$

$g(h(x)) = 2x - 15.$

Step 3: Substitute the result into f(x).

$f(g(h(x))) = -1(2x - 15)$

Step 4: Expand and simplify.

$f(g(h(x))) = -2x + 15.$

S4

For f(x) = -x + 7, find f(6x - 6).

Answer:

$-6x + 13$

Working:

Step 1: Treat 6x - 6 as one complete input and substitute it for x.

$f(6x - 6) = -1(6x - 6) + 7$

Step 2: Expand the bracket.

$f(6x - 6) = -6x + 6 + 7$

Step 3: Collect the constant terms.

$f(6x - 6) = -6x + 13$

Teacher answers & working

Function Notation, Composite and Inverse Functions · Expression Inputs and Composition

ALG-2F88-L2 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

E1

For $f(x) = x^2 - x - 3$, find $f(-2x - 3)$.

Answer:

$4x^2 + 14x + 9$

Working:

Step 1: Substitute the whole expression -2x - 3 wherever x appears.

$f(-2x - 3) = 1(-2x - 3)^2 - 1(-2x - 3) - 3$

Step 2: Expand the squared bracket and the linear bracket.

$f(-2x - 3) = 4x^2 + 12x + 9 - 1(-2x - 3) - 3$

Step 3: Collect like powers of x.

$f(-2x - 3) = 4x^2 + 14x + 9$

E2

For $f(x) = x + 3$, $g(x) = 5x - 2$, and $h(x) = 3x + 4$, find $f(g(h(x)))$.

Answer:

$15x + 21$

Working:

Step 1: Start with the innermost function, h(x).

$h(x) = 3x + 4$

Step 2: Substitute h(x) into g(x).

$g(h(x)) = 5(3x + 4) - 2$

$g(h(x)) = 15x + 18.$

Step 3: Substitute the result into f(x).

$f(g(h(x))) = 1(15x + 18) + 3$

Step 4: Expand and simplify.

$f(g(h(x))) = 15x + 21.$

Teacher answers & working

Function Notation, Composite and Inverse Functions · Expression Inputs and Composition

ALG-2F88-L2 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

E3

For f(x) = -3x - 8, find f(-2x + 6).

Answer:

6x - 26

Working:

Step 1: Treat -2x + 6 as one complete input and substitute it for x.

$f(-2x + 6) = -3(-2x + 6) - 8$

Step 2: Expand the bracket.

$f(-2x + 6) = 6x - 18 - 8$

Step 3: Collect the constant terms.

$f(-2x + 6) = 6x - 26$